

Andrew Willoughby

Duke University, Department of Biology and Salk Institute for Biological Studies
e: andrew.willoughby@duke.edu | w: andrewwilloughb.github.io

CURRENT POSITION

NSF Postdoctoral Research Fellow – Duke University, Durham, NC 2024-present

Project: Developing *Streptocarpus* as a model of meristem formation and plant regeneration |

Advisor: Lucia Strader and John Willis

EDUCATION and TRAINING

UNC Chapel Hill, Department of Biology—Chapel Hill, NC Ph.D. in Biology: 2018-2024 |

Advisor: Prof. Zachary Nimchuk

Dissertation: [CLE Signaling in Root Development and Its Cytoplasmic Requirements](#), defended 3/22/2024

University of Oklahoma, Department of Microbiology and Plant Biology —Norman, OK

B.S. in Plant Biology with Honors: 2014-2018 | **Advisor:** Sharon Kessler and Ben Holt

Boyce Thompson Institute at Cornell REU, Ithaca, NY 2016| **Advisor:** Jim Giovannoni

GRANTS + AWARDS (\$412,250 total)

2025 Modelling Plant Stem Cells: Evolution, Development, and Regeneration Workshop – Early Career Researcher delegate, Company of Biologists

2024 Community Science Program Annual Call – Department of Energy Joint Genome Institute
→**Gesneriaceae Genomes to Unlock Regeneration:** Genomics and Transcriptomes for 6 Gesneriaceae species

2024 Honorable Mention poster award – New Phytologist Next Generation Scientists Symposium

2024 New Phytologist Next Generation Scientists delegate – \$100

2024 NSF Postdoctoral Research Fellowship in Biology (PRFB, PGRP) – \$240,000

2023 Gilbert Travel Award – \$750, UNC Department of Biology

2022 Alma Holland Beers Scholarship – \$5,500, UNC Department of Biology

2019 Alma Holland Beers Scholarship – \$5,500, UNC Department of Biology

2019 W.C. Coker Fellowship – \$11,500, UNC Department of Biology

2019 Doctoral Merit Assistantship – \$2,500, UNC Chapel Hill Graduate School

2017 Honors Undergraduate Research Assistantship – \$9,000, OU Honors College (Spring)

2017 Undergraduate Research Opportunities Program – \$5,900, OU Office of Undergraduate Research

2017 Honors Undergraduate Research Assistantship – \$9,000, OU Honors College (Fall)

2017 Mentored Research Fund Award – \$5,000, OU Office of Undergraduate Research

2017 Goldwater Scholarship Honorable Mention

2016 Research Experience for Undergraduates – \$5,250, Boyce Thompson Institute at Cornell

2016 George L. and Cleo S. Cross Endowed Scholarship – \$1,000, OU Dept Micro and Plant Biology

2014 National Merit Scholarship Award – \$116,200, University of Oklahoma

PUBLICATIONS

14. **Andrew C. Willoughby**, Isabelle M. Trier, Hang Xue, Hongwei Jing, Adam Seroka, Irene Yuan, Samantha K. Powers, Ryan J. Emenecker, Elizabeth M. Frick, Tara A. Enders, and Lucia C. Strader. Plant UBX-Domain Containing Protein 2 (PUX2) promotes peroxisome homeostasis by regulating PEROXIN5 turnover (in preparation for submission in late 2026)

13. Van Etten J[#], Han S[#], Burns JA, Lhee D, **Willoughby A**, Stephens TG, Chille E, Benites LF, Sleith RS, Bhattacharya D, Yoon HS. “Crawling under the radar: two novel *Paulinella* species expand knowledge about the ecology and evolution of a primary plastid-containing amoeba lineage”. *Journal of Phycology*.
[#]equal contribution **Large media attention including in the [NYTimes](#)**

12. Zhiming Ma, Haolan Xu, Eden Vina Grate, Junbao Ma, **Andrew Willoughby**, Choon-Peng Chng, Chia-Heng Hsiung, Zhu Qiao, Deqian Zong, Ruoyu Jiang, Bozeng Tang, Liam Holt, Changjin Huang, Yong-

- Gui Gao, Zin Zhang, Lucia Strader, Endao Han, Yansong Miao. "Tissue-Scale Gradients of Macromolecular Crowding Orchestrate Plant Biomolecular Condensation" (*In revision at Nature*)
11. **Andrew C. Willoughby.** (2025). More extraordinary model systems for regeneration. 'Meristematic leaves in *Streptocarpus*: unlocking regeneration'. *Development*, 152(20), p.dev205215.
 10. Daniel S. Jones, Reid Selby, Pedro Jimenez-Sandoval, **Andrew C. Willoughby**, Emily Yaklich, Anna T. DiBattista, Jakub Baczynski, Feng Wang, Teng Zhang, Vandana Gurung, Ashley D. Crook, Andra-Octavia Roman, Erika Moore-Pollard, Riley Schuld, Jennifer R. Mandel, Paula Elomaa, John M. Burke, Julia Santiago, Zachary L. Nimchuk. Floral innovation through modifications in stem cell peptide signaling. *bioRxiv* (2025) 2025.06.27.661788; 10.1101/2025.06.27.661788 (**in revision at Science**)
 9. Smith, E.S., John, A., **Willoughby, A.C.**, Jones, D.S., Galvão, V.C., Fankhauser, C. and Nimchuk, Z.L., 2025. Canalization of flower production across thermal environments requires Florigen and CLAVATA signaling. *Current Biology* (2025) 10.1016/j.cub.2025.06.001
 8. Gensel, Patricia G., Amelia Milano, **Andrew Willoughby**, and James Belcher. "A New Zosterophyll with Novel Emergence and Cuticle Features from the Early Devonian of New Brunswick, Canada." *International Journal of Plant Sciences* 186, no. 3 (2025): 152-166.
 7. Thomas A DeFalco[#], Anne Pauline[#], Sean R. James[#], **Andrew Willoughby**, Oliver Johannndrees, Yasmine Genolet, Anne-Marie Pullen, Cyril Zipfel, Christian S. Hardtke, and Zachary L. Nimchuk. "A conserved regulatory module regulates receptor kinase signaling in immunity and development." *Nature Plants* (2022). [#]equal contribution
 6. **Andrew C. Willoughby**, and Zachary L. Nimchuk. "WOX going on: CLE peptides in plant development." *Current Opinion in Plant Biology* 63 (2021): 102056
 5. **Andrew C. Willoughby**[#], Ashley D. Crook[#], Ora Hazak, Satoshi Okuda, Kylie R. VanDerMolen, Cara L. Soyars, Pietro Cattaneo, Natalie M. Clark, Rosangela Sozani, Michael Hothorn, Christian S. Hardtke, and Zachary L. Nimchuk. "BAM1/2 receptor kinase signaling drives CLE peptide-mediated formative cell divisions in *Arabidopsis* roots." *Proceedings of the National Academy of Sciences* 117, no. 51 (2020): 32750-32756. [#]equal contribution
 4. Burr, Christian A., Jinjing Sun, Maria V. Yamburenko, **Andrew C. Willoughby**, Charles Hodgins, Samantha Louise Boeshore, Augustus Elmore et al. "The HK5 and HK6 cytokinin receptors mediate diverse developmental pathways in rice." *Development* 147, no. 20 (2020): dev191734.
 3. Jones, Daniel S., Xunliang Liu, **Andrew C. Willoughby**, Benjamin E. Smith, Ravishankar Palanivelu, and Sharon A. Kessler. "Cellular distribution of secretory pathway markers in the haploid synergid cells of *Arabidopsis thaliana*." *The Plant Journal* 94, no. 1 (2018): 192-202.
 2. Davis, Thomas C., Daniel S. Jones, Arianna J. Dino, Nicholas I. Cejda, Jing Yuan, **Andrew C. Willoughby**, and Sharon A. Kessler. "*Arabidopsis thaliana* MLO genes are expressed in discrete domains during reproductive development." *Plant Reproduction* 30, no. 4 (2017): 185-195.
 1. Jones, Daniel S., Jing Yuan, Benjamin E. Smith, **Andrew C. Willoughby**, Emily L. Kumimoto, and Sharon A. Kessler. "Mildew Resistance Locus o function in pollen tube reception is linked to its oligomerization and subcellular distribution." *Plant physiology* 175, no. 1 (2017): 172-185.

EDITORIAL PUBLICATIONS

- Willoughby, Andrew C.**, Aberrant RNA identification reveals triggers of transgene silencing *The Plant Cell* (2025) (In Brief article as Assistant Features Editor)
- Willoughby, Andrew C.**, Illuminating growth: Celebrating the life and legacy of Dr. Joanne Chory *Plant Physiology* (2025) (In Brief article as Assistant Features Editor)
- Willoughby, Andrew C.**, It's a small world: *Sinningia* double flower cultivars share the same *GLOBOSA1* allele *The Plant Cell* (2025) (In Brief article as Assistant Features Editor)
- Willoughby, Andrew C.**, and Lucia C. Strader. "Apical hook opening of plant seedlings: Unfolding the role of auxin and the cell wall." *Developmental Cell* 59.24 (2024): 3194-3196. (invited Preview article)
- Willoughby, Andrew C.** "Illuminating the future: Enhanced glowing plants achieved by rewiring metabolism." *The Plant Cell* (2024) (In Brief article as Assistant Features Editor)
- Willoughby, Andrew C.** "Air plant genomes shed light on photosynthesis innovation." *The Plant Cell* (2024) (In Brief article as Assistant Features Editor)
- Willoughby, Andrew C.** "Rapid strawberry domestication left room to grow." *The Plant Cell* (2024) (In Brief article as Assistant Features Editor)
- Willoughby, Andrew C.** "CRISPR, chimerism, and chromothripsis: A technique for studying DNA repair in

plants." *The Plant Cell* (2023): 3916-3917. (In Brief article as Assistant Features Editor)
Willoughby, Andrew C. "Mind the pretzels and the pint: The PUB keeps you thirsty." *The Plant Cell* (2023): 3639-3640. (In Brief article as Assistant Features Editor)

TEACHING

1. **Guest lectures** —Intro to Astronomy—Duke—Short lecture on primary endosymbiosis and the origins of complex life- Fall 2026
2. **Guest lectures**— Advanced Genetics —Duke Department of Biology—Two part lecture on molecular genetics and developmental biology- Spring 2025
3. **Guest lectures**— Food and Culture —OU Honors College— Genetic Engineering and Food- Fall 2022, Spring 2018, Spring 2017
4. **Guest lecture**— Seminar in Biology —UNC Chapel Hill Department of Biology— Meristems in *Marchantia* and Alternation of Generation- Fall 2021
5. **Lab Instructor/Teaching Assistant**— Introduction to Botany Lab —UNC Chapel Hill Department of Biology— Responsible for prep, content, and instruction of two lab sections, updated course content before semester and again due to COVID-19 pandemic interrupting semester - Spring 2020
6. **Teaching Assistant**— First Year Research Experience Program —OU Honors College— Spring 2018
7. **Teaching Assistant**— Practical Bioinformatics —OU Dept Microbiology and Plant Biology— Spring 2018
8. **Private Tutor**—high school, and college Math, Biology, and ACT prep – 2014-2018

TALKS

1. **Willoughby AC**, Strader L (August 2026) Developing *Streptocarpus* as a Model for Regeneration. Botany 2026 Tucson, AZ. Invited talk.
2. **Willoughby AC**, (July 2026) Freaks of Nature: What Makes *Streptocarpus* So Special? Gesneriad Society Convention 2026 Wheeling, IL. Invited talk.
3. **Willoughby AC**, Strader L (May 2026) Levering *Streptocarpus* to Study Regeneration Recalcitrance. Triangle Plant Molecular Biology meeting, Durham NC. Flash talk.
4. **Willoughby AC**, Strader L (May 2025) Developing *Streptocarpus* as a Model for Regeneration. University of Edinburgh. Edinburgh, Scotland. Invited talk.
5. **Willoughby AC**, Strader L (May 2025) Developing *Streptocarpus* as a model system for plant regeneration. Modelling Plant Stem Cells. Company of Biologists Workshop, one of ten accepted early career applicants. Ware, England.
6. **Willoughby AC**, Strader L (March 2025) Developing *Streptocarpus* as a model system for plant regeneration. Iowa State Crop Bioengineering Center. Invited talk.
7. **Willoughby AC**, Strader L (November 2024) A New Model System for Plant Regeneration. Duke Regeneration Center. Durham, NC. Invited talk.
8. **Willoughby AC**, Willis, J, Strader L (September 2024) *Streptocarpus* as a meristem model. Plant Genome Research Program annual meeting, NSF, Alexandria, VA. Flash talk. **Willoughby AC**, Willis, J, Strader L (September 2024) *Streptocarpus* as a meristem model. Plant Genome Research Program annual meeting, NSF, Alexandria, VA. Flash talk.
9. **Willoughby AC**, Strader L (September 2023) Developing *Streptocarpus* as a model system for plant regeneration. Climate-Plant Innovation Center Symposium, Duke University, Durham, NC. Invited talk.
10. **Willoughby AC**, Nimchuk Z (September 2020) What can we learn about CLE signaling from *bam* roots? Biology Department Trainee Seminar, UNC Chapel Hill, Chapel Hill, NC
11. **Willoughby AC**, Crook A, Nimchuk Z (June 2020) CLE peptide ligands and BAM receptors control asymmetric cell division in root ground tissue. American Society of Plant Biologists Conference 2020. Online.
12. **Willoughby AC**, Nimchuk Z (May 2019) Plant signaling in development and patterning. Biology Department Seminar, UNC Chapel Hill, Chapel Hill, NC
13. **Willoughby AC**, Holt BF (Spring 2017). The Role of *NF-Y* and *HY5* in Floral Development. Undergraduate Research Day Symposium, University of Oklahoma, Norman, OK
14. **Willoughby AC**, Holt BF (Fall 2017). The Role of *NF-Y* and *HY5* in Floral Development. Oklahoma Academy of Sciences, Oklahoma Microscopy Society Technical Meeting. Rogers State University, Claremore, OK.
15. **Willoughby AC**, Holt BF (Spring 2017). The Role of *NF-Y* and *HY5* in Floral Development. Undergraduate Research Day Symposium, University of Oklahoma, Norman, OK
16. **Willoughby AC**, Wang A, Giovannoni J (Summer 2016) High Pigment Mutations and Environmental Interactions. Scientists of the Future Symposium, Cornell University, Ithaca, NY

POSTER PRESENTATIONS

1. **Willoughby AC**, Strader L (October 2025) Developing *Streptocarpus* as a model system for plant regeneration. Salk Science at the Seaside. Salk Institute for Biological Studies. La Jolla, CA
2. **Willoughby AC**, Strader L (July 2025) Developing *Streptocarpus* as a model system for plant regeneration. Plant Biology 2025, Milwaukee, WI.
3. **Willoughby AC**, Strader L (March 2025) Developing *Streptocarpus* as a model system for plant regeneration. ASPB Southern Section, NC State University, Raleigh, NC
4. **Willoughby AC**, Strader L (October 2024) Developing *Streptocarpus* as a model system for plant regeneration. Triangle Regeneration Biology Symposium, Duke University, Durham, NC
5. **Willoughby AC**, Willis, J, Strader L (September 2024) Developing *Streptocarpus* as a model system for plant regeneration. Plant Genome Research Program annual meeting, NSF, Alexandria, VA
6. ***Willoughby AC**, Strader L (June 2024) Developing *Streptocarpus* as a model system for plant regeneration. New Phytologist Next Generation Scientists Symposium, Duke University, Durham, NC *Honorable mention poster award
7. **Willoughby AC**, Nimchuk Z (August 2023) The Role of *PUB4* in CLE signaling. American Society of Plant Biologists Conference 2023, Savannah, GA.
8. **Willoughby AC**, Crook A, Nimchuk Z (June 2021) Control of cell cycle and formative divisions by an extracellular CLE peptide. FASEB Mechanisms of Plant Development Conference 2021, Online.
9. **Willoughby AC**, Crook A, Nimchuk Z (June 2021) Control of cell cycle and formative divisions by an extracellular CLE peptide. International Conference for Arabidopsis Research 2021, Online.
10. **Willoughby AC**, Kieber, J (November 2018) *HK5* and *HK6* control development in Rice. BBSP Poster Session, UNC Chapel Hill, Chapel Hill, NC
11. **Willoughby AC**, Holt BF (Summer 2017). The Role of *NF-Y* and *HY5* in Floral Development. International Conference on Arabidopsis Research 2017, St. Louis, MO
12. **Willoughby AC**, Holt BF (Summer 2017). The Role of *NF-Y* and *HY5* in Floral Development. Curiosity to Creativity Symposium, University of Oklahoma, Norman, OK
13. **Willoughby AC**, Jones DS, Kessler SA (Spring 2015). Characterization of MLO6, a 7-Transmembrane Protein in *Arabidopsis thaliana*. Honors First Year Research Experience Symposium, University of Oklahoma, Norman, OK

SERVICE TO THE PROFESSION

The Plant Cell Assistant Features Editor, May 2023- January 2026

PNAS Journal Club Panelist, October 2022- January 2024

PROFESSIONAL MEMBERSHIP

1. Botanical Society of America —2025-present
2. Gesneriad Society —2023-present
3. American Society for Plant Biologists—2020-present
4. American Society for Plant Biologists Southern Section—2020-present

TECHNICAL SKILLS

- 🔧 Microscopy: Brightfield, Confocal, SEM
- 🔧 Sample preparation and optimization for light and electron microscopy
- 🔧 High molecular weight DNA extraction, chromatin confirmation capture
- 🔧 Genome assembly
- 🔧 Plant cultivation and basic microbiology
- 🔧 Plant tissue culture and *Agrobacterium*-mediated transformation of nonmodel plants
- 🔧 Primer design, PCR, gel electrophoresis
- 🔧 RNA-sequencing experimental design, sampling, and analysis
- 🔧 Immunocytochemical antibody staining, western blot and imaging
- 🔧 Proficient in Linux/Command Line and pretty good in R